

EXHIBIT G

UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.

Before the Honorable Clark S. Cheney
Chief Administrative Law Judge

In the Matter of

CERTAIN SEMICONDUCTOR DEVICES,
PRODUCTS CONTAINING SAME, AND
COMPONENTS THEREOF

Investigation No. 337-TA-1500

RESPONSIVE DECLARATION OF DR. SANJAY K. BANERJEE
REGARDING CLAIM CONSTRUCTION

I, Sanjay K. Banerjee, Ph.D., declare as follows:

I. INTRODUCTION AND QUALIFICATIONS

1. I incorporate by reference herein §§ I, II, III, IV, V, VI, VII, and VIII of my opening declaration regarding claim construction, which I understand was submitted on August 25, 2026.

2. I have been asked by Respondents to respond to Complainant's Opening Claim Construction Brief and the declaration of Dr. David Liu¹ submitted therewith regarding disputed terms in U.S. Patent Nos. 9,865,546 ("546 patent") and 10,707,167 ("167 patent"), which are asserted in the above-captioned investigation. I understand that Complainant has asserted infringement by Respondents of all claims (1-6) of the '167 patent and all claims (1-14) of the '546 patent.

II. MATERIALS CONSIDERED

3. In arriving at my opinions in this declaration, I have considered all materials referenced in § VII of my opening declaration regarding claim construction, submitted on August

¹ Dr. Liu's declaration was submitted as Exhibit 1 to Complainant's Opening Claim Construction Brief.

25, 2026, and the documents and information cited and discussed herein, including the exhibits listed below.

Exhibit No.	Description
N/A	Complainant's Opening Claim Construction Brief of August 25, 2026, and all exhibits cited therein and attached thereto

III. RESPONSIVE OPINIONS REGARDING CLAIM CONSTRUCTION

A. “layer” / “first liner layer” / “refractory metal layer”

Disputed Claim Term	Complainant's Proposed Construction	Respondents' Proposed Construction
“layer” / “first liner layer” / “refractory metal layer” ('546 patent, claims 1-14; '167 patent, claims 1, 3, and 4)	Plain and ordinary meaning; not indefinite.	Indefinite or, if not indefinite: “a separate and defined [first liner / refractory metal] coating of material”

4. I disagree with Dr. Liu's statement that “[t]he '546 and '167 claims use the term [‘layer’] consistently with its well-known usage in the art.” Liu Decl., ¶ 34. As discussed in my opening declaration, the '546 and '167 patents' claims use the term “layer” in ways that provide no reasonably certain scope to a POSA.

5. It is insufficient that the '546 and '167 patents' claims provide certain requirements about the relative location of the recited layers and their relationships with other claimed structures. These requirements do not remedy the fact that a POSA, applying Complainant's vague interpretation of the “layer” terms, would not know which coating(s) of material(s) could be grouped together or subdivided and labeled as “layers” for purposes of the claims.

6. For example, Dr. Liu points out that the patents' specification “teaches that . . . a ‘layer’ may be a ‘bilayer.’” Liu Decl., ¶ 35 (citing '546 patent, 7:3-8). But Dr. Liu does not

explain when or why a POSA should consider two coatings of materials to be a bilayer instead of two separate layers.

7. In addition, Dr. Liu does not provide any opinion regarding when or why a POSA should consider a single coating of material to be part of two different “layers.” [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

8. Dr. Liu cites what he purports to be a webpage of Respondent Tower that uses the term “layer.” Liu Decl., ¶ 35 (citing Ex. 6 at 1). But the content of this exhibit refers to separate and defined coatings of materials, such as Cu or Al, consistent with the plain-and-ordinary meaning of the term “layer” as reflected in Respondents’ alternative proposed construction. As such, this exhibit is irrelevant to Complainant’s indefinite use of the term “layer,” which departs from this plain-and-ordinary meaning for the reasons provided in my opening declaration.

9. I disagree with Dr. Liu’s statement that “a POSA would not understand how ‘separate’ a coating must be from other parts in order to be sufficiently ‘defined.’” Liu Decl., ¶ 37. To the contrary, I believe that a POSA would understand the term “layer” to necessarily refer to a coating of material that is separate and defined. Otherwise, the POSA would be unable to coherently identify or describe what a “layer” is, as reflected in Complainant’s Infringement and Domestic Industry Contentions.

10. I also disagree with Dr. Liu’s statement that a “layer,” in the context of the ’546 and ’167 patents, is not necessarily a “coating of material.” Liu Decl., ¶ 37. The ’167 patent’s and ’546 patent’s asserted claims require a “dielectric layer over the semiconductor substrate,” meaning that the dielectric layer is a coating of material over the semiconductor substrate. The

“first liner layer” and “refractory metal layer” in the ’167 patent’s claims are expressly described as respectively “coating” a sidewall of the contact opening and the first liner layer. Similarly, the claims of the ’546 patent expressly recite the first liner layer as “conformally coat[ing]” the contact opening and the refractory metal layer as being “over the first intermix region,” meaning that the first liner and refractory metal layers are also coatings of materials in the ’546 patent. All embodiments in the patents depict every layer as a coating of material, and a POSA would not have understood the claims of the ’167 and ’546 patents to cover any “layer” that is not a coating of material. Indeed, Dr. Liu provides no example of any “layer” that would *not* be a coating of material.

11. I disagree with Dr. Liu’s statement that Respondents’ alternative proposed construction would exclude any embodiment in the specification of the ’546 or ’167 patents. Respondents’ alternative proposed construction does not require a coating of material that is uniformly made up of only a single element or alloy. Liu Decl., ¶ 38. Instead, the material could comprise a bilayer of Ti/TiN for refractory metal purposes or a bilayer of different dielectric materials, so long as the larger “layer” is identifiable by its purpose such that it is separate and defined. *See* ’546 patent, 4:54-5:23 (describing potential bilayer of different dielectric materials, which the POSA would collectively recognize as being a “dielectric layer” because of the collective purpose of such layers to serve as an electrically insulating material whose main purpose is to control electric fields while preventing unwanted current flow); *id.*, 6:64-7:23 (describing a bilayer of Ti/TiN that would have been known to serve adhesion (Ti) and barrier (TiN) purposes in the contact stack). In other words, the “material” referenced in Respondents’ alternative proposed construction need not be a uniform thickness of a single element or alloy. It could be made up of different elements or alloys, so long as the identifiable “material” of the layer is

separate and defined, such that a POSA could distinguish it from other layers or components. Dr. Liu does not explain how a POSA could understand the scope of the claims with reasonable certainty if the claimed layers are not separate and defined.

12. Dr. Liu asserts that Respondents' alternative proposed construction excludes the embodiment of a first liner layer that partially diffuses into an intermix region. Liu Decl., ¶ 38. I disagree. The first liner layer is a separate and defined coating of material in both the '546 and '167 patents' claims. The '167 patent requires an intermix region "having a composition including a material used in the first liner layer." '167 patent, claim 1. This does not mean that the first liner layer itself, as claimed, is inside the intermix region. A POSA would have understood the term "intermix region" to describe a separate component of the claimed semiconductor structure that includes *material used* in the first liner layer, not the first liner layer itself (which is recited as being a separate and defined coating of material). Likewise, the '546 patent's claims recite method steps, one of which recites either "depositing a first liner layer to conformally coat the contact opening" in claim 1 and its dependent claims or "depositing a first liner layer to conformally coat each contact opening" in claim 8 and its dependent claims. A POSA would understand this separate and defined coating of material to be a first liner layer. The fact that a subsequent method step recites creation of a "first intermix region" that includes a portion of the first liner layer does not change the fact that a POSA would have understood the first liner layer, as originally deposited, to be a separate and defined coating of material.

13. For the reasons herein and in my opening declaration, it is my opinion that the terms "layer," "first liner layer," and "refractory metal layer" should be construed to mean "a separate and defined [first liner / refractory metal] coating of material" or, alternatively, that they are

indefinite for failure to provide reasonable certainty about the scope of the claimed inventions to a POSA.

B. “intermix region”

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“intermix region” (’546 patent, claims 1, 2, 6, 8, 9, and 13; ’167 patent, claims 1, 2, and 6)	Plain and ordinary meaning; not indefinite.	Indefinite

14. I disagree with Dr. Liu’s opinion that the term “intermix region” is not indefinite. Dr. Liu provides no opinions regarding how much intermixing is required for materials to qualify as an “intermix region.” It is not enough that the claims specify certain materials that must be included in the intermix region or where the intermix region is located. A POSA would not understand how much intermixing is required for a “region” to qualify as an “intermix region” in light of the intrinsic record.

15. For the reasons herein and in my opening declaration, it is my opinion that the term “intermix region” is indefinite for failure to provide reasonable certainty about the scope of the claimed inventions to a POSA.

16. I reserve the right to supplement or amend this declaration to address any evidence, opinion, statement, or position taken by Complainant or any expert who submits a declaration in support of Complainant’s claim construction positions.

17. I swear under penalty of perjury under the laws of the United States that the facts set forth herein are true and correct to the best of my knowledge. I am willing to testify as to the truth and accuracy of the foregoing at any hearing in the above-captioned Investigation.

Dated: September 3, 2026

Signed:

A handwritten signature in blue ink, reading "Sanjay Banerjee", is positioned above a horizontal line. The signature is written in a cursive style.

Sanjay K. Banerjee, Ph.D.